Heart of Algebra

ANSWER KEY

Solving linear equations

1 Solve for x: $4x - 9 = 23$.

$4x = 32$, so $x = 8$.

2 Solve for x: $\frac{3x+2}{5} = 4$.

Multiply both sides by 5: $3x + 2 = 20$, so $3x = 18$, giving $x = 6$.

3 Solve for x: $6(x - 3) = 2(x + 9)$.

Expand: $6x - 18 = 2x + 18$. Subtract $2x$: $4x - 18 = 18$, so $4x = 36$, giving $x = 9$.

Linear inequalities

4 Solve $5x - 3 > 12$.

$5x > 15$, so $x > 3$.

5 Solve $-2x + 7 \le 15$.

Subtract 7: $-2x \le 8$. Divide by -2 and flip the inequality: $x \ge -4$.

6 Solve $3(x - 2) \ge x + 4$.

Expand: $3x - 6 \ge x + 4$. Subtract x: $2x - 6 \ge 4$, so $2x \ge 10$, giving $x \ge 5$.

Slope and equations of lines

7 Find the slope of the line through $(2, 5)$ and $(6, 13)$.

$m = \frac{13-5}{6-2} = \frac{8}{4} = 2$.

8 Write the equation, in slope-intercept form, of the line with slope -3 passing through $(0, 7)$.

$y = -3x + 7$.

9 Find the slope of a line perpendicular to $y = \frac{2}{5}x - 1$.

A perpendicular slope is the negative reciprocal of $\frac{2}{5}$, which is $-\frac{5}{2}$.

Systems of equations by substitution

10 Solve the system: $y = 3x - 4$ and $2x + y = 11$.

Substitute: $2x + (3x - 4) = 11$, so $5x - 4 = 11$, giving $5x = 15$, $x = 3$. Then $y = 3(3) - 4 = 5$. Solution: $(3, 5)$.

11 Solve the system: $x - 2y = 1$ and $3x + y = 17$.

From the first equation, $x = 1 + 2y$. Substitute: $3(1 + 2y) + y = 17$, so $3 + 6y + y = 17$, giving $7y = 14$, $y = 2$. Then $x = 1 + 2(2) = 5$. Solution: $(5, 2)$.

Systems of equations by elimination

12 Solve the system: $3x + 2y = 19$ and $x - 2y = 1$, using elimination.

Adding the two equations eliminates y: $4x = 20$, so $x = 5$. Substituting into $x - 2y = 1$: $5 - 2y = 1$, so $2y = 4$, giving $y = 2$. Solution: $(5, 2)$.

13 Solve the system: $4x - y = 10$ and $2x + y = 14$, using elimination.

Adding the two equations eliminates y: $6x = 24$, so $x = 4$. Substituting into $2x + y = 14$: $8 + y = 14$, so $y = 6$. Solution: $(4, 6)$.

Absolute value equations

14 Solve $|2x - 5| = 9$.

$2x - 5 = 9 \Rightarrow x = 7$, or $2x - 5 = -9 \Rightarrow x = -2$. Solutions: $x = 7$ or $x = -2$.

15 Solve $|x + 4| = 10$.

$x + 4 = 10 \Rightarrow x = 6$, or $x + 4 = -10 \Rightarrow x = -14$. Solutions: $x = 6$ or $x = -14$.

Interpreting linear models

16 A phone plan costs $C = 20 + 0.05m$ dollars for m minutes of calls. Interpret the meaning of the numbers 20 and 0.05 in this context.

The 20 represents a fixed monthly fee (the cost when $m = 0$ minutes are used), and the 0.05 represents the cost per minute — each additional minute of calls adds 0.05 dollars to the bill.

17 Company A charges $C = 30 + 0.20d$ and Company B charges $C = 0.35d$ for a d-mile trip. Find the mileage at which both companies charge the same amount, and state which company is cheaper for a 150-mile trip.

Set equal: $30 + 0.2d = 0.35d$, so $30 = 0.15d$, giving $d = 200$ miles. At 150 miles: Company A charges $30 + 0.2(150) = 60$ dollars; Company B charges $0.35(150) = 52.5$ dollars. Company B is cheaper at 150 miles.

Systems with no solution or infinitely many solutions

18 For what value of k does the equation $6x + 4 = kx + 4$ have infinitely many solutions?

For infinitely many solutions the equation must hold for every x, so the coefficients of x must match: $k = 6$ (the constants already agree, $4 = 4$).

19 Explain why the system $2x - 3y = 6$ and $4x - 6y = 10$ has no solution.

Multiplying the first equation by 2 gives $4x - 6y = 12$. This contradicts the second equation, $4x - 6y = 10$ — the same left-hand expression cannot equal two different values, so the lines are parallel and never intersect.

Compound inequalities

20 Solve the compound inequality $-3 < 2x + 1 \leq 9$.

Subtract 1 from all three parts: $-4 < 2x \leq 8$. Divide by 2: $-2 < x \leq 4$.

21 Solve $4 - 3x \geq 13$.

Subtract 4: $-3x \geq 9$. Divide by -3 and flip the inequality: $x \leq -3$.

Multi-step word problems

22 Two numbers have a sum of 50 and a difference of 12. Find the two numbers.

Let the numbers be x and y with $x + y = 50$ and $x - y = 12$. Adding: $2x = 62$, so $x = 31$. Then $y = 50 - 31 = 19$. The numbers are 31 and 19.

23 The sum of three times a number and 7 equals 5 less than twice the number. Find the number.

$3x + 7 = 2x - 5$. Subtract $2x$: $x + 7 = -5$, so $x = -12$.

A well-designed word problem: ticket sales at a community theater

24 This question has three parts. A community theater sells adult tickets for 15 dollars and child tickets for 8 dollars. On Saturday the theater sold 120 tickets total and collected 1,450 dollars. (a) Set up a system of two equations for the number of adult tickets a and child tickets c sold. (b) Solve the system to find how many of each ticket type were sold. (c) On Sunday the theater raises the adult price to 18 dollars but keeps the child price the same. If the same number of adult and child tickets sell as on Saturday, find the new total revenue.

(a) $a + c = 120$ and $15a + 8c = 1450$. (b) From the first equation, $a = 120 - c$. Substitute: $15(120 - c) + 8c = 1450$, so $1800 - 15c + 8c = 1450$, giving $1800 - 7c = 1450$, so $7c = 350$, $c = 50$. Then $a = 120 - 50 = 70$. So 70 adult tickets and 50 child tickets were sold. (c) New revenue $= 18(70) + 8(50) = 1260 + 400 = 1660$ dollars.